

Intelligent Sensor Placement for Hot Server Detection in Data Centers

Xiaodong Wang, Xiaorui Wang, *Member, IEEE*, Guoliang Xing, *Member, IEEE*, Jinzhu Chen, Cheng-Xian Lin, and Yixin Chen, *Senior Member, IEEE*

Abstract—Recent studies have shown that a significant portion of the total energy consumption of many data centers is caused by the inefficient operation of their cooling systems. Without effective thermal monitoring with accurate location information, the cooling systems often use unnecessarily low temperature set points to overcool the entire room, resulting in excessive energy consumption. Sensor network technology has recently been adopted for data-center thermal monitoring because of its nonintrusive nature for the already complex data center facilities and robustness to instantaneous CPU or disk activities. However, existing solutions place sensors in a simplistic way without considering the thermal dynamics in data centers, resulting in unnecessarily degraded hot server detection probability. In this paper, we first formulate the problems of sensor placement for hot server detection in a data center as constrained optimization problems in two different scenarios. We then propose a novel placement scheme based on computational fluid dynamics (CFD) to take various factors, such as cooling systems and server layout, as inputs to analyze the thermal conditions of the data center. Based on the CFD analysis in various server overheating scenarios, we apply data fusion and advanced optimization techniques to find a near-optimal sensor placement solution, such that the probability of detecting hot servers is significantly improved. Our empirical results in a real server room demonstrate the detection performance of our placement solution. Extensive simulation results in a large-scale data center with 32 racks also show that the proposed solution outperforms several commonly used placement solutions in terms of detection probability.

Index Terms—Data centers, servers, sensor placement, thermal monitoring, computational fluid dynamics, power management

1 INTRODUCTION

POWER and thermal management has become a key challenge in the design of large-scale data centers. In a 2007 report to the US Congress [1], the Environmental Protection Agency estimated that the annual data center energy consumption in the US will grow to over 100 billion kWh at a cost of \$7.4 billion by 2011. One of the key reasons for data centers to have excessive energy consumption is the inefficient operation of their cooling systems (e.g., a set of computer room air conditionings (CRAC)), which can account for up to half of their energy consumption [1]. To reduce the energy consumption caused by excessive cooling, a variety of thermal control schemes have been recently proposed. For example, HP researchers developed an adaptive vent tile technology [2] to automatically adjust mechanical louvers mounted to the floor vent tiles so that detected hot areas can get more cool air. Bash et al. [3]

proposed to adaptively adjust the temperature set point and air flow rate (i.e., fan speed) of each individual CRAC in a data center. Other control algorithms (e.g., [4], [5]) have also been proposed to handle detected hot servers by throttling their CPU frequencies, migrating away their workloads, or shutting them down if necessary. For all the aforementioned schemes to work in practice, effective thermal monitoring mechanisms are needed to ensure that the locations of hot areas and servers can be accurately detected and the information can be sent to the controllers promptly. Without effective thermal monitoring with accurate location information, data center operators have to excessively overcool the entire room [6], leading to the waste of cooling energy.

In addition to improving cooling energy efficiency, accurate thermal monitoring is more critical to the thermal reliability of a data center. Overheating servers in a data center commonly have significantly shortened lifetimes, as the lifetime of an electronic component decreases with the increase of its working temperature [7], [8]. More importantly, undesired high temperatures in a data center also increase the probability of thermal emergency, leading to the shutdown of various IT equipment and the disruption of service availability. For example, the most used online encyclopedia, Wikipedia, went down on July 5th, 2010, because the failure of a cooling unit caused server overheating and then the power outage of the data center [9]. In this Wikipedia example, if the hot areas created by the failed cooling unit had been more accurately detected, the blackout could have been avoided by effectively directing the air flows of remaining CRAC units to cool down the overheating servers.

Efficient thermal monitoring is challenging, given the data centers' complex air flow and thermal dynamics. Traditionally, simple thermostats or wired temperature

- X. Wang and X. Wang are with the Department of Electrical and Computer Engineering, The Ohio State University, 2015 Neil Avenue, Columbus, OH 43210. E-mail: {wangxi, xwang}@ece.osu.edu.
- G. Xing and J. Chen are with the Department of Computer Science and Engineering, Michigan State University, 3115 Engineering Building, East Lansing, MI 48824-1226. E-mail: {glxing, chenjinz}@msu.edu.
- C.-X. Lin is with the Department of Mechanical and Materials Engineering, Florida International University, 10555 W. Flagler St, EC3445, Miami, FL 33174. E-mail: lincx@fiu.edu.
- Y. Chen is with the Computer Science and Engineering, Washington University in St. Louis, Campus Box 1045, One Brookings Drive, St. Louis, MO 63130. E-mail: chen@cse.wustl.edu.

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probes are used to provide coarse-grained thermal monitoring, which cannot effectively support the monitoring granularity required by the thermal control schemes to conduct energy-efficient cooling. Wireless sensor network (WSN) technology has recently been identified as an ideal candidate for data-center thermal monitoring [10], [3] due to several of its salient advantages. First, it can provide good coverage with accurate localization for *global* thermal management decisions in a data center. Second, it is nonintrusive, as the sensors use wireless communications and thus require no additional network and facility infrastructure in an already complicated data-center environment.

Compared with data center thermal monitoring solutions based on motherboard sensor readings, using additional WSN technology also has promising advantages. First, compared to the thermal sensors on motherboards, the wireless sensors are less sensitive to instantaneous CPU or disk activities, leading to less noisy thermal readings [10]. Second, the sensors on server motherboards are currently working in isolation and thus cannot provide an overall thermal picture of the data center, which is important for system level data center thermal management. Finally, low-end sensors used on server motherboards commonly have noises and hardware biases that may lead to undesirable detection performance. Recent studies [11], [12] have shown that the collaborative data fusion of multiple sensors can significantly improve the detection accuracy.

Although WSN technology has shown promise in data-center thermal monitoring, an important issue that has been overlooked by existing solutions is how to optimally place sensors in a data center such that all the possible overheating locations are well covered and monitored with maximized detection probabilities. Currently, many real data centers just simply place the same number of sensors on each rack uniformly at a constant distance from each other, without considering the thermal dynamics in the data center. For example, in a real data center located in HP Labs in Palo Alto, five sensors are placed on the front side of each rack from the top to the bottom to keep the inlet temperature at or below 24 °C for all running servers [4]. Five sensors are used for each rack because it is usually preferable not to put too many wireless sensors on a rack for the considerations of space and cost, due to the very dense installation of high-density servers (e.g., up to 128 blade servers per rack). In addition, a highly dense deployment of sensors may cause the wireless network to have significantly increased levels of channel contention and thus, unacceptably long communication delays [13], [14]. However, such a simplistic sensor placement strategy may result in an unnecessarily degraded detection probability. In contrast, an optimized placement solution can intelligently place sensors based on the systematic analysis of the thermal dynamics in the data center, by considering the locations of the CRAC systems and the server racks, as well as the rack layout and various air flows in the room. As a result, better coverage can be achieved for servers that have a greater potential to become overheating. Consequently, given the same number of sensors, such an optimized solution can lead to a significantly improved detection probability and so a better chance for the existing control schemes to prevent thermal emergencies.

In this paper, we propose a novel sensor placement scheme for improved hot server detection performance, which can enhance the thermal control operations in data centers. Our placement scheme is developed based on the numerical results from computational fluid dynamics (CFD), a powerful mechanical fluid dynamic analysis approach. CFD is widely used to analyze the fluid dynamics in various engineering fields, such as aircraft engine design and environmental analysis for buildings. CFD has already been used by data center designers to make intelligent decisions on layout design and rack deployments, but *not yet* for sensor placements. In this paper, we use CFD to model the thermal environment of a given data center under different thermal emergency conditions and apply interpolation techniques to improve the thermal analysis results from CFD. We seek to solve two sensor placement problems that address the overheating server detection performance under two different conditions. The first problem is when the number of sensors is given, we seek to place all the given sensors in the data center so that the potential overheating servers (due to workload increases, CRAC failures, and so on.) at any location can be detected with the maximum detection probability. In the second problem, considering the still high cost of each wireless temperature sensor, we seek to minimize the number of sensors needed to achieve a required overheating server detection probability. We formulate these two problems as constrained optimization problems based on data fusion techniques to allow sensors to make collaborative detection decisions of overheating servers. Based on the formulation and the CFD analysis, we design heuristic algorithms to find near-optimal placement solutions with a significantly reduced computational complexity, despite a huge search space. We evaluate our sensor placement approach both on a testbed in a server room with 13 server racks and more than 100 servers and in simulation with a CFD model of a large-scale data center.

Specifically, the contributions of this paper are fourfold:

- While the current WSN-based thermal monitoring solutions in many real data centers rely on simplistic sensor placement without considering the thermal dynamics in the data center, we propose a novel sensor placement scheme to intelligently place sensors for maximized hot server detection probabilities.
- We propose to use CFD to model the thermal dynamics of a data center in various overheating scenarios (e.g., different servers are overheating due to workload increases or CRAC failures). CFD analysis provides a theoretical foundation for our sensor placement solution. We apply interpolation techniques to further refine the numerical results from CFD.
- We formulate optimal sensor placement as two constrained optimization problems under different user requirements and propose heuristic algorithms to find near-optimal solutions with a significantly reduced computational complexity.
- We evaluate our sensor placement scheme in a real server room with 13 racks and more than 100

servers. Two CFD models with different granularity are established for our testbed. A CFD model for a typical large-scale data center with 32 racks and 640 servers is also established for simulation evaluation. Both our empirical and extensive simulation results demonstrate that our placement solution can significantly improve hot server detection performance, compared with different commonly used baseline placement schemes.

The remainder of this paper is organized as follows: Section 2 highlights the distinction of our work by discussing related work. Section 3 presents the data fusion model we used and the formulations of the two hot server detection problems in data centers. Section 4 introduces the fundamentals of the CFD approach and provides an example of how to model a server room in CFD. Section 5 elaborates on how to use the numerical results from CFD in our sensor placement problems and proposes heuristic algorithms to solve the problems. In Section 6, we evaluate our sensor placement scheme using simulations. In Section 7, we present results from the empirical experiment on our testbed. Section 8 concludes the paper and discusses the possible future work.

2 RELATED WORK

Thermal management in data centers has been widely studied in the past. Moore et al. [5] have proposed a temperature-aware workload placement scheme for data center. Optimization schemes for data center thermal management using model-based approaches have been proposed in [15], [16]. An automated, online, predictive thermal management scheme for data centers is also proposed in [17]. However, none of the above-mentioned studies has explored the possibility of using WSNs. Several projects have adopted sensor networks in data center for temperature monitoring. For example, a hybrid wired and WSN is used in [10] for data center thermal monitoring. A sensor network is used in [3] to manipulate conventional CRAC units within the data center. Li et al. [18] proposed a temperature prediction approach for data center based on sensor readings. However, none of the past methods addresses how to intelligently deploy sensors to improve the hot server detection performance. Our work is different from all the aforementioned research. We not only explore the benefits of using WSNs for temperature monitoring in data centers, but also maximize the detection probability of the potential overheating servers. It is important to note that our sensor placement solution is complementary to existing thermal control scheme (such as [2], [3], [4], [5]) because more accurate thermal monitoring can significantly enhance the performance of thermal control. For example, as shown in [3], even a very simplistic deployment of temperature sensors can lead to a 50 percent saving in cooling energy.

CFD has been used to model the data center operating environment and server rack operating conditions. Patel et al. [19] have used CFD to model and analyze the air temperature specification in the data center. Impact of CRAC failures on static provisioning has also been studied using CFD models in [20]. Jeohwang et al. [21] have modeled the thermal profile for an operating rack in detail

in to provide a bridge between the individual component thermal status and data center thermal profile. Different from all the previously mentioned studies, our paper uses CFD to model different thermal emergency situations in data centers when servers (or racks) are overheating at any possible locations. We then use the CFD modeling results to guide the sensor deployment for the various overheating conditions, such that any thermal emergency associated with server workload dynamics or CRAC failures can be effectively monitored and reported by the sensor network.

Target detection and monitoring is one of the most important tasks of WSNs. Several existing projects have explored how to deploy sensors effectively to improve the detection and monitoring performance. A sensor placement scheme based on the multivariate Gaussian process model is proposed in [22], which provides most informative results after the data training period. A fast sensor placement approach for fusion-based target detection is proposed in [11], [12] to minimize the number of deployed sensors while achieving assured detection performance. Different from these previous schemes of sensor deployment, the sensor deployment approach we propose leverages on the computational results from CFD which analyzes the thermal condition of a monitored field based on theoretical thermal dynamics. Furthermore, the model training approach proposed in [22] is not applicable for data center thermal emergency monitoring, because the thermal emergency scenario should not be created simply for the collection of training data.

3 HOT SERVER DETECTION PROBLEM

In this section, we first introduce the hot server (hot spot) detection model in sensor networks. We then formally formulate two data center hot server detection problems, i.e., the detection probability maximization problem and the sensor number minimization problem, respectively.

3.1 Hot Server Detection Model

Temperature sensors are usually manufactured with a variation on their sensing accuracy. To conduct data center temperature monitoring and hot server detection with a reasonable confidence of the sensing results, sensor nodes should collaborate with each other when making detection decisions. Data fusion [23], [24], [25], a widely adopted technique for improving the detection performance of sensor systems by collaboration, is well suited in this scenario.

It is clear that temperatures at the locations far from a heat source are less likely to be correlated with this source. Therefore, we define a *fusion region* of each monitored location as a sphere with a *fusion radius* R , where the monitored location is the center of that sphere. The fusion radius is a parameter that characterizes a sensor monitoring system. For each different sensor monitoring system, the fusion radius can be different and needs to be tuned to reach the best data fusion result. The sensors within the fusion region of a monitored location should collaborate to make the detection decision for that location. We assume all the sensors in our monitoring system are identical, which means all sensors have the same sensor reading error distributions. Therefore, we adopt a simple data fusion

scheme that calculates the average temperature from all the sensors within the fusion region of the monitored spot and compare the average value with a detection threshold η . If the average temperature is larger than the threshold, the decision of a hot server detection is positive.

The measurements of a sensor are usually corrupted by noise. Denote the measurement noise strength measured by sensor i as N_i , which follows the zero-mean normal distribution with a variance of σ^2 , i.e., $N_i \sim \mathcal{N}(0, \sigma^2)$. Since the measurement of a signal by a sensor is in the energy form, which in our case is the heat amount the sensor sensed, a noise item is also in energy form. The noise energy should be added to the final temperature reading. Therefore, the final measured temperature, T_m , from a sensor at location (x_i, y_i, z_i) can be presented as

$$T_m(x_i, y_i, z_i) = T_r(x_i, y_i, z_i) + N_i^2, \quad (1)$$

where T_r is the real temperature at that location without noise.

If we assume that there are n sensors within the data fusion region of a monitored spot and the measurement noise is Gaussian Noise, i.e., $N_i/\sigma \sim \mathcal{N}(0, 1)$, the detection probability of the hot server existence and the false alarm rate that a false overheating detection is reported can be calculated as

$$P_D = 1 - \chi_n \left(\frac{n\eta - \sum_{i=1}^n T_r(x_i, y_i, z_i)}{\sigma^2} \right) \quad (2)$$

$$P_F = 1 - \chi_n \left(\frac{n(\eta - C)}{\sigma^2} \right), \quad (3)$$

where η is the detection threshold of overheating with noise and C denotes the real temperature threshold without noise. $\chi_n(\cdot)$ in the above equations denotes the Chi-square distribution [26]. The detail derivation of (2) and (3) can be found in the supplementary file of this paper, which can be found on the Computer Society Digital Library at <http://doi.ieeecomputersociety.org/10.1109/TPDS.2012.254>.

3.2 Detection Probability Maximization

We now formulate the detection probability maximization problem. We assume that there are M locations (e.g., blade servers) in the data center room for which we need to monitor temperature. It is always desirable to cover as many locations as possible in the thermal monitoring of a data center. For example, we may want to monitor the temperature at the inlet or outlet of each rack or even each server. Therefore, the number of monitored locations in a large data center is usually large. It is therefore unrealistic to place an individual sensor for every monitored location, especially in a WSN environment. It is not only because of the space limitation and cost, but also because of the resulting high network density. A very dense wireless sensor placement often leads to undesired network communication quality because of severe interferences and channel contentions among wireless transmissions [10], [13].

Given a limited and reasonable number of sensors, $N < M$, we need to find the placement of these N sensors such that we can detect the overheating emergency at any of the M locations with the highest possible confidence. Before

presenting our formal formulation, we first introduce the following notation:

- l_i , monitored location i with location coordinates (x_{li}, y_{li}, z_{li}) .
- P_{Fi} , the false alarm rate of reporting an overheating emergency at l_i .
- P_{Di} , the detection probability of an overheating emergency at l_i .
- n_i , the sensor number within the fusion region of l_i .
- (x_{sj}, y_{sj}, z_{sj}) , the sensor placement location for sensor s_j within the fusion region of l_i .

Our goal is to maximize the average detection probability of all the monitored locations

$$\arg \max_{(x_{sj}, y_{sj}, z_{sj}) \forall j} \frac{1}{M} \sum_{i=1}^M P_{Di} \quad (4)$$

subject to the following constraint:

$$P_{Fi} \leq \alpha \quad \forall 1 \leq i \leq M, \quad (5)$$

where α is the detection false alarm rate requirement. We note that the false alarm rate needs to be bounded in many practical scenarios. Without constraining the false alarm rate, a system can report false detection results, which can lead to the waste of the processing resources, such as human efforts and computing resources. This formulation of detection probability maximization is equivalent to minimize the misdetection of the monitoring system, including both missing a real overheating target and detecting a false overheating target (false alarm incident). Note that network connectivity can be another constraint to be considered in our formulation to ensure that every sensor is connected to the network for effective data fusion. However, due to the high-density installation of servers (e.g., blade servers) in the data center, our experience shows that connectivity is not a major concern, because the communication range of a wireless sensor is 30 m indoors [27].

For a certain sensor placement, $P_{Fi} \leq \alpha$ is a necessary condition in our problem. By (3), we convert the constraint in (5) to $\eta_i \geq \frac{\sigma^2 \chi_n^{-1}(1-\alpha)}{n_i} + C$, a constraint for the detection threshold η at monitored location i , where $\chi_n^{-1}(\cdot)$ is the inverse function of $\chi(\cdot)$. Using this equation, we can obtain the desired detection threshold based on the required false alarm rate and use it to calculate the detection probability. From (2), we know that P_{Di} decreases when η_i increases. Therefore, to maximize the detection probability, we remove the inequality in the constraint and only use the lower bound α . Hence, η_i can be calculated as

$$\eta_i = \frac{\sigma^2 \chi_n^{-1}(1-\alpha)}{n_i} + C. \quad (6)$$

3.3 Sensor Number Minimization

Another interesting problem in data center thermal monitoring is minimizing the number of placed sensors. The cost for a wireless temperature sensor, such as a Telosb Mote, is still high today, which makes it cost forbidden to place a large amount of sensors for data center hot server monitoring, as the number of locations that need to be

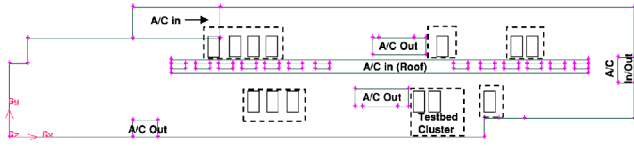


Fig. 1. Top view layout of the server room in the EECS department at the University of Tennessee. The size is about $30 \text{ m} \times 7 \text{ m} \times 3.4 \text{ m}$. Small solid boxes represent server racks and the dotted boxes represent rack clusters.

monitored can be high. Moreover, the associated installation cost often increases drastically with the number of sensors. Minimizing the number of wireless sensors can effectively reduce the cost of using WSNs to conduct data center thermal monitoring. To achieve the goal of sensor number minimization, we can extend the probability maximization problem in last section by changing the detection probability objective as a constraint. Assume that we have M locations in the data center room whose temperatures need to be monitored, we need to minimize the sensor number N that can yield a detection probability higher than β and a false alarm rate lower than α at each monitored location. Formally, the problem can be formulated as follows:

$$\arg \min_{(x_{sj}, y_{sj}, z_{sj}) \forall j} N \quad (7)$$

subject to the following constraints:

$$P_{F_i}(S_N) \leq \alpha \quad \forall 1 \leq i \leq M \quad (8)$$

$$P_{D_i}(S_N) \geq \beta \quad \forall 1 \leq i \leq M, \quad (9)$$

where S_N is the list of locations of all the N sensors.

4 CFD MODELING FOR DATA CENTER

In this section, we introduce CFD, the tool we use to analyze the thermal conditions in a data center. We also provide examples of how to model a server room in practice using Fluent [28], a widely used CFD modeling software package, with a simplified CFD server rack model and a finer-grained CFD server rack model.

4.1 CFD Modeling and Example

CFD is a fluid mechanics approach that analyzes problems of fluid flows based on numerical methods and algorithms. The key for CFD modeling is to solve the governing transport equations, which are a set of coupled differential equations represented in the conservation law form. The detail equations can be found in the supplementary file, available online, of this paper. For a complicated environment, such as a data center, no closed-form solutions can be found for the airflow and heat transfer of the entire system. Therefore, the most fundamental consideration in CFD is how to treat a continuous fluid in a discretized fashion, such that numerical methods can be applied to find the solutions. Most CFD software packages apply the control volume method to find numerical solutions. In our project, we use Gambit and Fluent, two widely used CFD software packages from ANSYS, Inc., to perform the spatial domain meshing and solution finding, respectively.

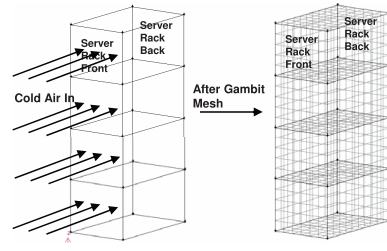


Fig. 2. Example of meshing a server rack in Gambit.

Fig. 1 shows an geometry model of a server room established in Gambit. This is the top view of the CFD geometry model for the server room in the EECS department at the University of Tennessee, Knoxville. There are 13 racks in the room, which is cooled by four closed-loop liquid-air HEX. The cold air is coming from three directions, which are the north west corner, the east side of the room and an overhead cold air duct on the ceiling. The hot air is recirculated back to the four AC out units in the room. To simplify the complexity of the server room CFD modeling, we group the servers in each rack into four blocks from top to bottom by equal physical size, as shown in Fig. 2. This approach was also adopted by other projects, such as in [19], to simplify the geometric establishments in CFD modeling. We will present a finer-grained CFD server rack model in next section. To apply the control volume method for numerical calculations, the geometry model needs to be meshed and discretized. Fig. 2 shows an example of using Gambit to perform the geometric mesh for a server rack. After the establishment of the discretized geometry model, we use Fluent to calculate the temperature distribution of the room. We adopt the k-epsilon turbulence model in Fluent. Although the server room we use in this example is not in a typical cold-isle-hot-isle server room setting, the k-epsilon turbulence model in CFD fully considers the air circulation among different points in the server room, such that any existing air circulation in the room can be modeled correctly. An example temperature distribution output of CFD simulation on the server room can be found in the supplementary file, available online, of this paper.

4.2 CFD Rack Model Improvement

We have introduced in the previous section a simplified rack model with four blocks in CFD to represent a single rack in our server room. This model simplifies the process of meshing the server room model during CFD modeling. It also reduces the time used for the CFD analysis by having more flexibility to control the mesh size in CFD modeling. However, as a tradeoff, it also has two problems. First, it reduces the accuracy of the thermal analysis in CFD since the thermal behavior of a simplified rack model can be different from a real rack. Second, we can only study the overheating scenarios for the monitored environment at the granularity of a single block.

In this section, we present a finer grained CFD server rack model that can offer a higher modeling accuracy at the cost of a higher computation complexity. The finer grained server rack model models each server on the server rack, instead of group a set of servers into a group. This new CFD server rack model allows us to simulate data center

overheating scenario at the granularity of a server. The downside of this finer grained rack model is that it takes more computing resource and time to conduct the overheating analysis by CFD software, since the mesh size is smaller than the previous rack model. Because of the limited computing resource (memory, and so on.) on the machine we run CFD analysis, we only use the finer grained CFD server rack model for two racks in our server room, which are the two racks indicated as testbed cluster in Fig. 1. A temperature map example of this testbed cluster can be found in the supplementary file, available online, of this paper.

5 CFD-GUIDED SENSOR PLACEMENT

In this section, we introduce how to use the results from the CFD analysis to guide sensor placement, targeting the two sensor placement problems formulated in previous sections. We also design heuristic algorithms for solving these two placement problems with a reduced searching space.

5.1 Overview of Our Approach

Using CFD tools for our sensor placement primarily involves two steps. In the first step, we establish a geometric model for the data center room in Gambit, mesh the geometry, and export the grid to Fluent. We then take measurements for the incoming cold air temperature and air flow rate from the inlet of every CRAC unit. These measurements, along with the power consumption of each block, are the input parameters to Fluent. There can be many various operating conditions for a data center. CFD simulation may not be able to cover all the different situations. However, the number of scenarios for a server to be overheating is actually limited. An overheating server can be caused by abnormally high inlet temperature, an unreasonably high workload, a failed fan, or the combination of these factors. Regardless of the specific type of overheating scenarios, the exhaust temperature in all these cases increases significantly. Therefore, we simulate the limited overheating scenarios for the sensor placement purpose to detect overheating servers. The power dissipation of each rack is set to the overheating scenario, one rack at a time, when solving the temperature distribution for each overheating scenario by an iterative solution procedure in Fluent. We monitor the exhaust temperature of each block such that the aforementioned overheating cases can be captured. For the detailed detection locations, we assume that our sensor placement needs to monitor the temperature of the center point at the back face of each block. This is a location close to the outlets of servers. The direction of the airflow at that location is mostly perpendicular to each server. Therefore, the temperature at the monitored point is mostly decided by the air convection from servers close to the point and the heat radiation from the surrounding area.

In the second step, we feed the results from the CFD analysis of all the overheating scenarios to our optimization algorithm to find the best locations for sensor placement. To solve the placement problem efficiently, we develop heuristic algorithms based on the constrained simulated annealing (CSA) approach [29]. Our sensor placement algorithm relies on the knowledge of temperature at any possible location in data center under overheating

condition. We use CFD result to get this knowledge. Based on the overheating temperature from CFD, which is governed by the physical laws of thermal dynamics, our algorithm optimizes the sensor locations to get a maximized detection probability. Since the results from the CFD analysis are temperatures at discretized locations, our algorithm features a spatial temperature interpolation approach [30] to interpolate the missing temperature values from the CFD results. By the spatial temperature interpolation, we can get the temperature value of any location in a data center room. The detailed interpolation process can be found in the supplementary file, available online, in this paper. The CFD results we feed to our algorithm is actually the interpolated CFD results.

Note that our approach can be easily modified to monitor the temperature difference between the inlet and the exhaust of a server, since the CFD simulation results also include all the inlet temperatures under different overheating conditions. We only need to apply the same approach to optimally place sensors for inlet temperature monitoring.

5.2 Lightweight Sensor Placement (LSP) Algorithms

Our goal is to find the optimal sensor placement locations that maximize the detection probability of all the overheating targets in the detection probability optimization problem, or minimizes the number of sensors to reach the required detection probability in the sensor number minimization problem. Since every sensor has three location coordinates, with N sensors to be placed in the domain, we need to solve a problem with $3N$ variables. A straightforward approach to this problem is to solve this entire problem at once. This can be achieved by a nonlinear programming solver based on the CSA algorithm [29].

CSA is an extension of the conventional simulated annealing algorithm for solving the global constrained optimization problem with discrete variables. Theoretically, CSA can reach a global optimal solution by converging asymptotically to a constrained global optimum with a probability of 1. However, a limitation of CSA is that its computational complexity grows exponentially with respect to the number of variables and the solution search space [29], [11]. The execution time of the algorithm can reach up to thousands of days with hundreds of sensors to place [11]. Therefore, it is not realistic to solve the entire placement problem as a whole. In this paper, we design a lightweight optimization algorithm based on CSA to reduce the complexity of the algorithm and find a near-optimal solution to our sensor placement problem.

Our LSP algorithm uses two techniques to speed up the solution search. From Fig. 1, we see that servers can be grouped based on their geographical locations. Based on the cluster pattern, our algorithm only searches possible sensor placement location within the vicinity of each cluster. Locations outside the fusion region of cluster boundary will not be searched; thus, the searching space is reduced. Second, our algorithm greedily adds sensor one by one to the solution set. A new sensor will be added to a cluster that can gain the maximum overall detection probability increasing. Note that during this step, only the cluster that has just adopted a sensor in the last round needs to be

recalculated for the sensor locations with a new sensor. All the other clusters have already been calculated for the sensor location solution with an additional sensor in previous rounds. By using the above search space reduction and greedy optimization techniques, the computational time can be significantly reduced. The pseudocode of the algorithm to solve the detection probability maximization problem, as well as a more detailed explanation of our algorithm can be found in the supplementary file, available online, of this paper.

6 SIMULATION RESULTS

In this section, we first explain the setup of our simulations. We then evaluate our sensor placement approach for the two sensor placement problems in simulations.

6.1 Experiment Overview

We conduct our simulation based on two different CFD models. The first model is for the server room shown in Fig. 1. The second model is for a large-scale data center model, which is introduced in detail in Section 6.2. For the first model, there are 13 racks in the server room. Each rack is divided into four blocks, with one monitored location at the center of the back side of each block, leading to 52 monitored locations in total. We use CFD to model the rack overheating scenario for all the 13 racks one by one. For each rack overheating scenario, we set the power dissipation of the overheating rack to 3,600 Watts per block, thus 14,400 Watts for a whole rack, which is the same parameter used in [19]. For other model inputs, including the cold air temperatures and flow rates from CRAC systems, we use the Tri-Sense digital temperature indicator from Cole-Power [31] to measure them. The CFD output data, the temperature of the entire room including the inlet and outlet of each server (block), for all 13 scenarios is collected as the input to the sensor placement algorithm. A comparison between the CFD temperature results and the real temperature measurement under a normal operation condition of the server room can be found in the supplementary file, available online, of this paper. The CFD results only slightly deviate from the real measurement. We further calibrate the data from CFD to compensate the system input error. After the calibration, the average temperature result discrepancy is reduced to 2 °C between CFD and real measurement.

The temperature data from CFD simulation is then used as input to the sensor placement algorithm and the sensor locations are calculated. We also get the temperature reading of each sensor location from the CFD data at the same time. Then, the sensor readings are fused together to report the temperature of each monitored location. The final detection probability is calculated as the percentage of all the overheating targets that is correctly reported by the placed sensors. In our analysis, we compare our sensor placement scheme, *CFD + LSP*, with two baseline approaches: *CFD + Proportional* and *Uniformly Random*. The two baselines differ from *CFD + LSP* in that they use different strategies to add sensors to clusters. In *Uniformly Random*, sensors are placed evenly into clusters. Within a cluster, the placement of sensors is random. In *CFD + Proportional*,

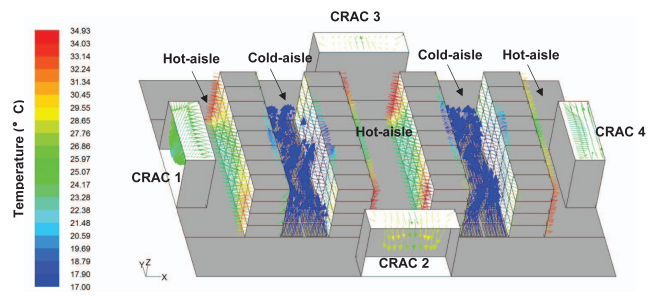


Fig. 3. CFD simulation results of a normal operation setting for a large-scale data center model with 32 racks in a typical cold-aisle hot-aisle arrangement. Air flows are colored based on their temperatures. The cold air (in blue) comes from the perforated tile in cold-aisle. The hot air comes out from servers and goes back to CRAC systems.

sensors are placed in the cluster based on cluster size. A cluster with a larger size gets more sensors. The placement in each cluster is derived from the CSA algorithm with the CFD results as input. Except otherwise stated in any specific experiments, for all the experiments in this section, we set the number of sensors to 10, the fusion range to 1 m, the temperature threshold to 35 °C and the false alarm rate to 5 percent.

6.2 Detection Probability Maximization on a Large-Scale Data Center Model

The probability maximization results based on the server room model introduced in Section 4.1 can be found in the supplementary file, available online, of this paper. In this section, we show the simulation results of the detection probability maximization problem with a large-scale raised-floor data center CFD model. Different from the CFD model of the server room where server racks are not put close to each other, the CFD model used in this experiment models a large-scale data center with densely deployed server racks. The model is shown in Fig. 3. It consists of 32 racks, arranged in four rows. Two cold aisles and three hot aisles are thus formed by this rack arrangement. This model is similar to the one used in [32] with an improvement on the model of the floor tiles. Instead of using fully open raised-floor tiles as in [32], we improve the model by using perforated floor tiles. As shown by Abdelmaksoud et al. [33] that a data center CFD model with perforated floor tiles can significantly improve the model accuracy.

Similar to the rack model introduced in Section 4.1, we model each rack in this large-scale data center with four stacked boxes. The heat dissipation of each box is set to 4,000 W. The temperature of the cold air from the four CRAC systems is set to 17 °C. We run CFD simulations with 16 different overheating scenarios. In each overheating scenario, we close one pair of perforated floor tiles (16 different pairs in 16 scenarios) to emulate a perforated tile failure in the data center. Eight of the total 16 scenarios are used as input to sensor placement algorithms. The validation of different sensor placement approaches is based on all the 16 overheating scenarios. The overheating temperature is set to 34 °C. When running our proposed *CFD + LSP* algorithm, the fusion range is set to 1 m in the experiments.

Fig. 4 shows the detection probability of all the targets in the 16 overheating scenarios with different numbers of

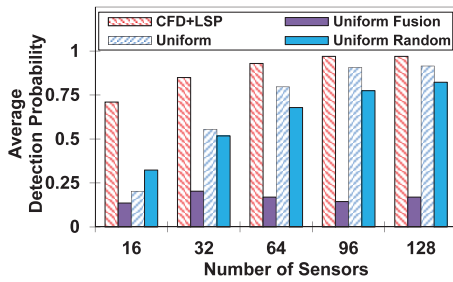


Fig. 4. Average detection probability on a large-scale data center CFD model in simulation.

sensors deployed. From the result, we see that $CFD + LSP$ has the best detection probability among all the sensor placement schemes. It has a detection probability higher than 80 percent when 16 sensors (i.e., 0.5 sensor per rack on average) are deployed. This is because $CFD + LSP$ places sensors at the expected overheating areas based on the results from the CFD overheating simulations. *Uniform Fusion* is the scheme placing sensors with even distance at the back of racks and employing data fusion to the readings from different sensors. We see that it performs the worst because it does not consider the overheating pattern when placing sensors, such that some sensors are placed at the places without overheating signature in any scenarios. As a result of those ineffective sensor placements, adopting data fusion to get the average temperature from multiple sensors in this scheme is misleading and thus degrades the detection performance even more. Without the misleading data fusion results, *Uniform* performs better than *Uniform Fusion*, but still worse than $CFD + LSP$. We further investigate the performance of the *Uniform Random* scheme, where in each evenly divided area, the sensors are placed randomly. Each data point of this scheme is the average result of 10 random placements. We see that it performs worse than *Uniform*, which shows that *Uniform* placement is relatively better than an arbitrary sensor placement using the *Uniform Random* scheme.

6.3 Sensor Number Minimization

In this section, we show the evaluation results of different placement schemes that solve the sensor number minimization problem. In this set of experiments, we do not show the performance of *Uniformly Random*. The reason is that *Uniformly Random* cannot reach the given detection probability requirement, which is set to larger than 70 percent in our experiments. This is because even with a large number of sensors, *Uniformly Random* randomly places all the sensors, some of which may be placed at the locations with low temperature signature. By including these sensors in the target temperature fusion, *Uniformly Random* may even degrade the overheating detection performance. Therefore, we only show the evaluation results of the $CFD + LSP$ and the $CFD + Proportional$ schemes.

Fig. 5 shows the results of the two different placement schemes when the overheating detection probability threshold is set to different values. We see that with a higher overheating detection probability requirement, more sensors are needed to reach the given requirement. Between the two placement schemes, $CFD + LSP$ always performs better than $CFD + Proportional$ by using fewer sensors.

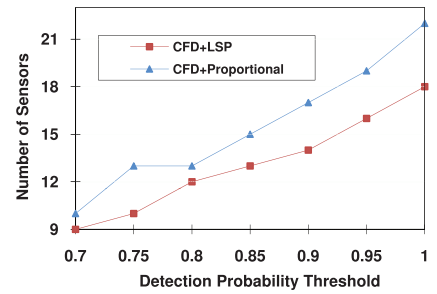


Fig. 5. Minimum number of sensors required to reach the given detection probability threshold.

placement within all the clusters together while $CFD + Proportional$ gives the different number of sensors proportionally to each cluster. The proportional assignment of sensors by $CFD + Proportional$ leads to unnecessary redundancy in some clusters, which is a waste of sensors. Fig. 6 shows the minimum number of sensors required when the scale increases from one cluster to six clusters. We see that with more clusters, thus larger size of problem, $CFD + LSP$ shows significantly better performance than $CFD + Proportional$ when minimizing the number of required sensors. The detection probability threshold in this set of experiments is set to 95 percent.

6.4 Results with Finer Grained CFD Model

In this section, we present simulation results based on the CFD analysis with the finer grained CFD server rack model introduced in Section 4.2. Since we only use the finer grained CFD server rack model for two racks, we only conduct experiments to place sensors on these two racks. We set half of the servers on these two racks to be overheating, as introduced in Section 4.2, which results in 35 overheating targets in total.

Before we can calculate the sensor placement positions, a reasonable fusion range needs to be decided. We run an experiment in CFD analysis by only setting one server to be overheating to evaluate the impact range of a single overheating server. We compare the temperature results from the CFD analysis for scenarios before and after setting the single server to be overheating. The results are shown in Fig. 7. We see that setting Server 17 to be overheating has different impact on the average temperature at the back of each server. The temperature at the back of Server 17 increases the most, while its neighboring servers, Servers 16 and 18, also show an increase in their temperatures. The temperatures at the servers further away from Server 17

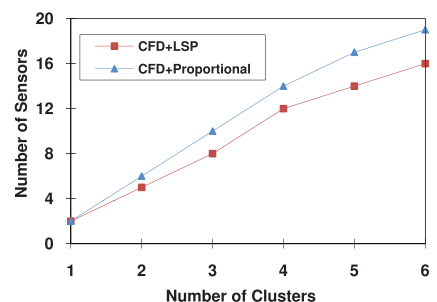


Fig. 6. Minimum number of sensors required for different number of clusters (the detection probability threshold is set to 95 percent).

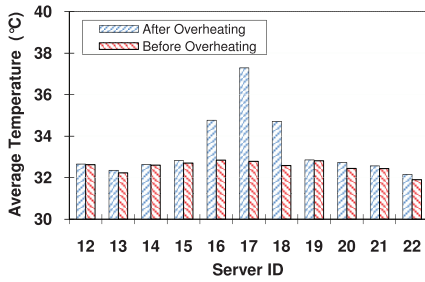


Fig. 7. CFD temperature results before and after Server 17 is set to be overheating.

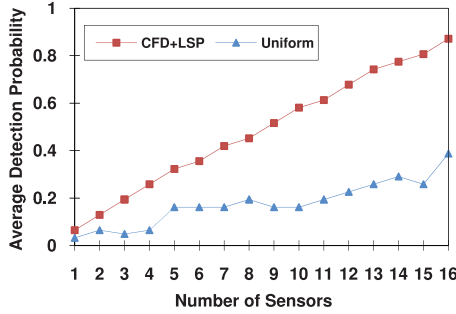


Fig. 8. Simulation result of the average detection probability with different numbers of sensors.

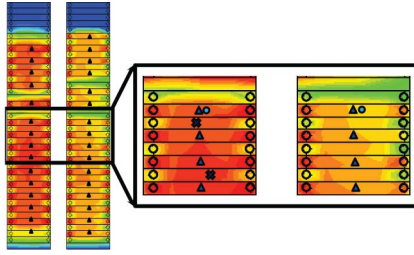


Fig. 9. Sensor placement result with two sensors. Circles represent the sensor placement of *Uniform*. Crosses represent the sensor placement of *CFD + LSP*.

show little difference before and after setting Server 17 to be overheating. Based on these results and the physical size (1U) of each server, we set the fusion range of our placement algorithm to 10 cm.

Fig. 8 shows the average detection probability with different number of sensors from our *CFD + LSP* approach and the *Uniform* approach in simulation. We see that *CFD + LSP* shows significantly better detection probability than the *Uniform* placement approach. The reason is that by using CFD to analyze the possible overheating scenarios and LSP algorithm to find the sensor placement locations, the probability of overheating detection is maximized. On the contrary, *Uniform* does not consider the overheating temperature signature and blindly places sensors evenly according to the space size and the number of sensors. Note that *Uniform* is the current practice used in many data centers (e.g., [4]). Figs. 9 and 10 show two placement examples by these two placement schemes. In Fig. 9, two sensors are used for overheating detection. *Uniform* places these two sensors evenly on the two racks. As a result, the *Uniform* placement only covers two overheating targets according to the fusion requirements. In contrast, *CFD + LSP* places the two sensors to cover four overheating servers. Fig. 10 enlarges a partial placement result of placing 10 sensors. We see that *Uniform* even places sensor at the

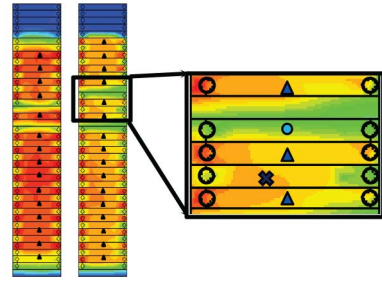


Fig. 10. Partial sensor placement result with ten sensors. 9 Circles represent the sensor placement of *Uniform*. Crosses represent the sensor placement of *CFD + LSP*.

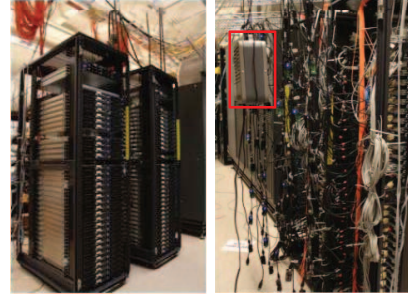


Fig. 11. Front and back sides of the server racks used in hardware experiments (the heater used to emulate overheating server block is highlighted).

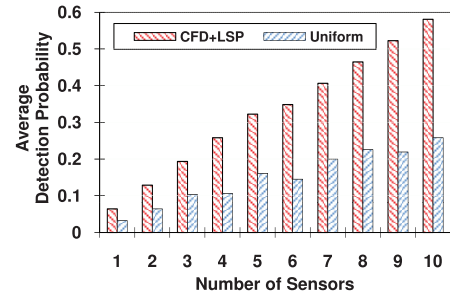


Fig. 12. Average detection probability on the hardware testbed with overheating at the inlet of each targeted server.

location of very low temperature (because no server is placed at that location), while *CFD + LSP* avoids that area and places sensor to cover more overheating targets.

7 HARDWARE TESTBED RESULTS

In this section, we show the experimental results from our hardware testbed in the server room based on the CFD analysis with the finer grained server rack model. Fig. 11 shows an image of the 2-rack hardware testbed we used in the experiment. When evaluating *CFD + LSP* approach, we extract the sensor placement locations from the simulation results and place sensors to the exact locations on the hardware testbed. We generate each of the 35 single server overheating scenarios separately by only warming up the inlet temperature of the overheating server using a hair dryer. After warming up each of the potential overheating server, we collect the sensor readings. Based on the reported temperature after data fusion, we evaluate whether the overheating is reported correctly. Each experiment is repeated five times. Note that for validation purpose, additional sensors can be put at each monitored location to measure the real temperature. The reported

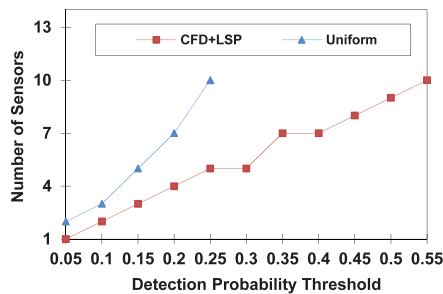


Fig. 13. Average detection probability on the hardware testbed with overheating at the inlet of each targeted server.

temperature from sensors placed by our approach can then be compared against the real temperature at those monitored locations.

Fig. 12 shows the overheating detection performance on the hardware testbed with different number of sensors. We see that the hardware experiment results confirm the simulation results that *CFD + LSP* can significantly increase the overheating detection probability. Fig. 13 shows the minimum number of sensors needed to reach a given average overheating detection probability. Because of the space limit on the two racks, we only place up to 10 sensors for the two different sensor placement schemes. We see that with the same average overheating detection probability requirement, *CFD + LSP* requires fewer sensors than *Uniform*. *Uniform* uses all the 10 sensors to reach an average overheating detection probability of 0.25, while *CFD + LSP* only needs five sensors to reach the same goal. With 10 sensors, *CFD + LSP* can reach an average overheating detection probability of 0.55. More testbed result based on the simplified rack model can be found in the supplementary file, available online, of this paper.

8 CONCLUSIONS

Efficient thermal monitoring is critical for today's data centers to significantly reduce cooling energy consumption and operating costs. WSN technology has recently been identified as an ideal candidate for data-center thermal monitoring. However, existing solutions adopted by real data centers place sensors in a simplistic way without considering the thermal dynamics in data centers, resulting in an unnecessarily degraded detection probability. In this paper, we have presented a novel sensor placement scheme for hot server detection in data centers based on CFD analysis of thermal dynamics, which uses both the cooling systems and servers as inputs in analyzing the thermal conditions of a given data center. Based on the CFD analysis, we apply data fusion and advanced optimization techniques to find the optimized sensor placement solution, such that the probability of detecting hot servers is maximized, or the number of required sensors to reach a desired hot server detection probability is minimized. Our solution features heuristic algorithms that significantly reduce the computational complexity of finding the near-optimal sensor placement scheme. Our empirical results on a hardware testbed and simulation results both demonstrate that our placement solution outperforms a commonly used uniform placement solution in terms of detection probability.

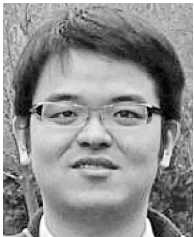
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Xiaodong Wang received the BS degree in electrical engineering from Shanghai Jiao Tong University, China, in 2006, and the MS degree in computer engineering from the University of Tennessee, Knoxville in 2009 and is currently working toward the PhD degree in the Department of Electrical and Computer Engineering at the Ohio State University. Before joining The Ohio State University, he was a PhD student at University of Tennessee, Knoxville. He received the first Min Kao Fellowship of Electrical Engineering and Computer Science Department at the University of Tennessee, Knoxville from 2007 to 2010. He also received the ESPN Graduate Student Fellowship and the Chancellors Award for Extraordinary Professional Promise Award from the University of Tennessee, Knoxville, in 2010 and 2011, respectively. In 2007, he worked at PDF Solutions, Inc., as a data analysis engineer.



Xiaorui Wang received the PhD degree from Washington University in St. Louis in 2006. He is an associate professor in the Department of Electrical and Computer Engineering at The Ohio State University. He received the US Office of Naval Research Young Investigator Award in 2011, the US National Science Foundation CAREER Award in 2009, the Power-Aware Computing Award from Microsoft Research in 2008, and the IBM Real-Time Innovation Award in 2007. He also received the Best Paper Award from the 29th IEEE Real-Time Systems Symposium in 2008. He is an author or coauthor of more than 60 refereed publications. From 2006 to 2011, he was an assistant professor at the University of Tennessee, Knoxville, where he received the EECS Early Career Development Award, the Chancellors Award for Professional Promise, and the College of Engineering Research Fellow Award in 2008, 2009, and 2010, respectively. In 2005, he worked at the IBM Austin Research Laboratory, designing power control algorithms for high-density computer servers. From 1998 to 2001, he was a senior software engineer and then a project manager at Huawei Technologies Co. Ltd., China, developing distributed management systems for optical networks. His research interests include power-aware computer systems and architecture, real-time embedded systems, and cyber-physical systems. He is a member of the IEEE and the IEEE Computer Society.



Guoliang Xing received the BS degree in electrical engineering and the MS degree in computer science from Xian Jiao Tong University, China, in 1998 and 2001, respectively, and the MS and DSc degrees in computer science and engineering from Washington University in St. Louis, in 2003 and 2006, respectively. He is an assistant professor in the Department of Computer Science and Engineering at Michigan State University. From 2006 to 2008, he was an assistant professor of computer science at the City University of Hong Kong. He received the National Science Foundation CAREER Award in 2010. He received the Best Paper Award at the 18th IEEE International Conference on Network Protocols in 2010. His research interests include wireless sensor networks, mobile systems, and cyber-physical systems. He is a member of the IEEE.



Jinzhu Chen received the BS degree in communication engineering from the University of Electronic Science and Technology of China, in 2005. From 2005 to 2009, he was an embedded software engineer at Delphi China Technical Center. He is currently working toward the PhD degree at Michigan State University. His research interests include wireless sensor networks and cyber-physical systems.



Cheng-Xian Lin received the PhD degree in mechanical engineering (thermal engineering) from Chongqing University, China. He is currently an associate professor in the Department of Mechanical and Material Engineering at FIU. His prior positions include associate professor in the University of Tennessee, Knoxville and Summer Faculty Fellow at Air Force Research Laboratory in WPAFB. He has authored and coauthored over 150 papers in peer-reviewed

journals and conference proceedings. His current research interests include computational fluid dynamics, heat transfer, thermal management, energy efficiency and renewable energy in built environments. He is a member of the ASME and ASHRAE.



Yixin Chen received the PhD degree in computing science from the University of Illinois at Urbana-Champaign in 2005. He is an associate professor of computer science at the Washington University in St Louis. His research interests include data mining, machine learning, artificial intelligence, optimization, and cyber-physical systems. He received the Best Paper Award at the AAAI Conference on Artificial Intelligence (2010) and International Conference

on Tools for AI (2005), and best paper nomination at the ACM KDD Conference (2009). His work on planning has received First Prizes in the International Planning Competitions (2004 and 2006). He has received an Early Career Principal Investigator Award from the Department of Energy (2006) and a Microsoft Research New Faculty Fellowship (2007). He is an associate editor for *ACM Transactions of Intelligent Systems and Technology* and *IEEE Transactions on Knowledge and Data Engineering*, and serves on the Editorial Board of *Journal of Artificial Intelligence Research*. He is a senior member of the IEEE.

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